

NAVIGATION

Hospital Overview

Patient Flow

Department Analytics

Resource Utilization

TOTAL ADMISSIONS

45,000

▲ 4.8%

ALOS

5.16 Days

▼ 0.3d

30-DAY READMISSION RATE

77.72%

▲ 2.1%

OCCUPANCY RATE

63.81%

▲ 1.4%

BED UTILIZATION

63.86%

▲ 1.1%

Total Admissions by Department

Axis: Dept → Admissions

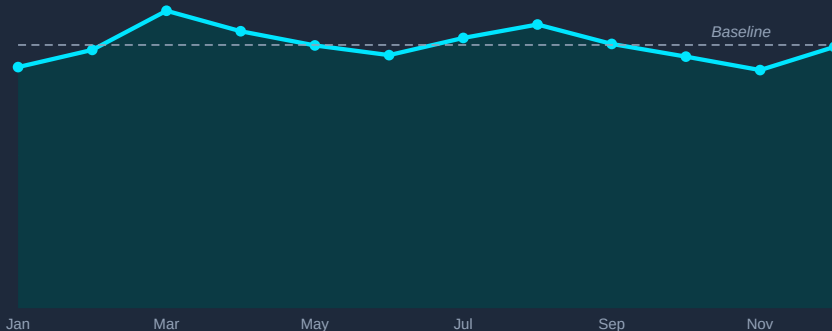
Horizontal Bar Chart



Monthly Admissions Trajectory Trend

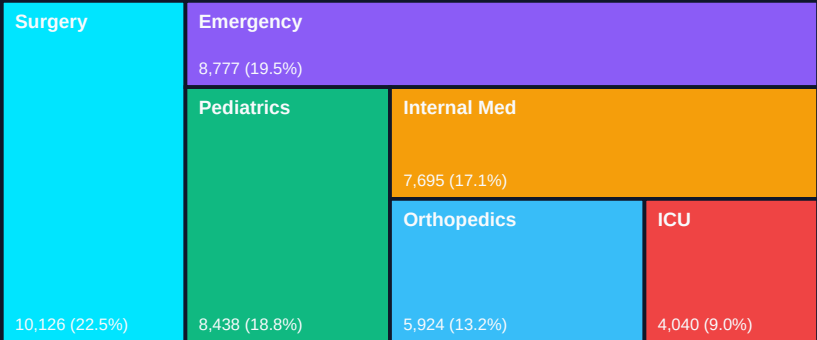
Axis: Month → Admissions

Continuous Line Chart



Admission Volume Treemap

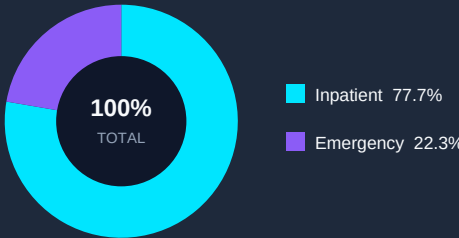
Proportional Department Distribution



Patient Admission Type Breakdown

Inpatient vs Emergency

Donut Chart



GLOBAL FILTERS

DEPARTMENT

All

PATIENT TYPE

All

DATE RANGE

2020-2025

MEDTRACK AI COPILOT & EXECUTIVE INSIGHT

AI

Top Admission Driver: Surgery (10,126)

Peak Month: March (3,874)

Capacity Strain: ICU (80.0%)

Surgery represents 22.5% of hospital admissions. ICU bed occupancy is at surge threshold (80.0%), warranting immediate capacity review.

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TOTAL INPATIENTS

34,973

▲ 3.2%

TOTAL EMERGENCY ADMISSIONS

10,027

▲ 1.9%

PEAK MONTH VOLUME

3,874

▲ March

AVERAGE STAY DURATION

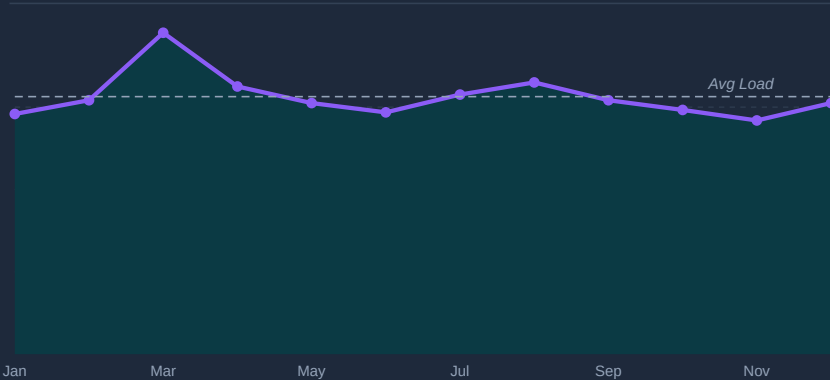
5.16 Days

▼ 0.2d

Peak Patient Load Trajectory by Month

Axis: Month → Intake

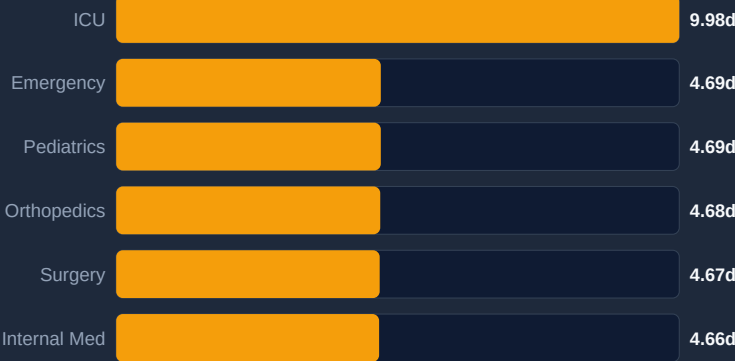
Monthly Intake Velocity



Average Length of Stay by Department

Axis: Dept → Days

Horizontal Bar Chart

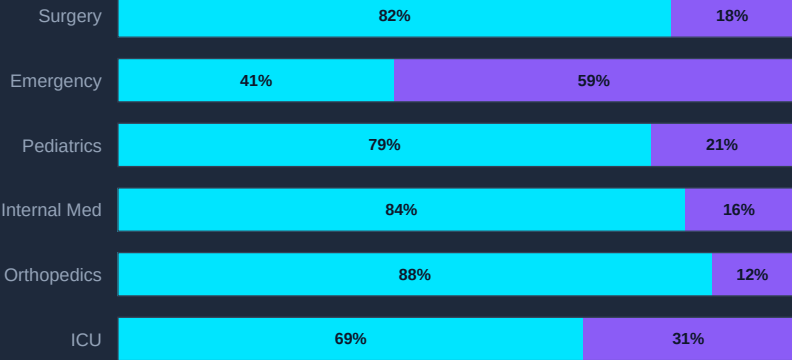


Department Admission Type Distribution

Inpatient vs Emergency

100% Stacked Bar Chart

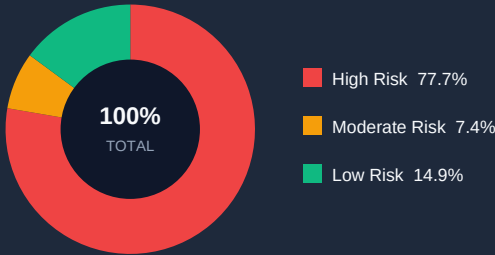
Inpatient Emergency



Readmission Risk Tier Classification

High / Moderate / Low

Donut Chart



Clinical Risk Stratification derived from 30-day chronological delta intervals computed in the Python data pipeline.

NAVIGATION

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DEPARTMENTS MONITORED

6

TOP EFFICIENCY DEPT ▲ Best

Surgery: 64.00

LOWEST EFFICIENCY DEPT ▼ Watch

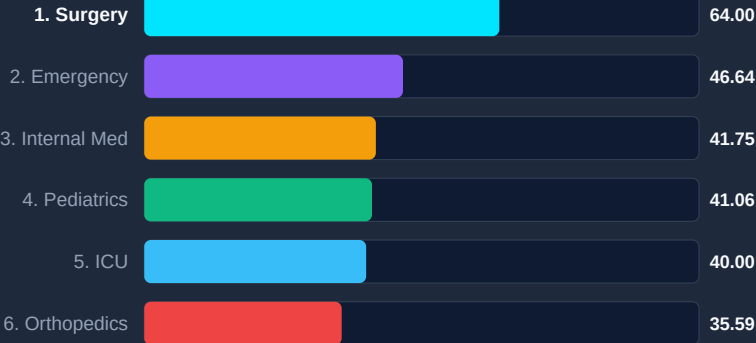
Orthopedics: 35.59

AVG WORKLOAD ▲ 6.4%

872.8 / Staff

Department Efficiency Score Leaderboard

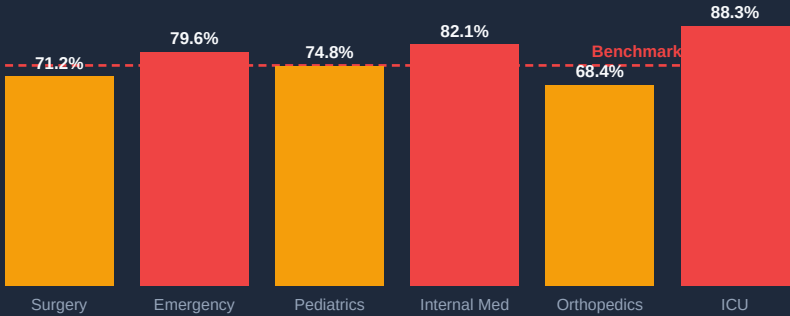
Ranked Progress Bars (0–100)



30-Day Readmission Rate vs. 75% Benchmark

Axis: Dept → Rate

Bar Chart



Department Performance Summary Matrix

Data Table — Cross-Departmental KPI Comparison

DEPARTMENT	TOTAL ADMISSIONS	AVG LOS	OCCUPANCY RATE	STAFF ALLOCATION	PATIENTS/STAFF	EFFICIENCY SCORE
Surgery	10,126	4.67d	68.2%	9	1,125.1	64.00
Emergency	8,777	4.69d	71.4%	8	1,097.1	46.64
Pediatrics	8,438	4.69d	59.8%	8	1,054.8	41.06
Internal Med	7,695	4.66d	61.0%	7	1,099.3	41.75
Orthopedics	5,924	4.68d	55.7%	8	740.5	35.59
ICU	4,040	9.98d	80.0%	7	577.1	40.00

GLOBAL FILTERS

DEPARTMENT

All

PATIENT TYPE

All

DATE RANGE

2020-2025



Interactive Cross-Filtering: Selecting any department isolates patient cohort metrics across all dashboards.

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Resource Utilization

TOTAL HOSPITAL BEDS

405

AVG OCCUPIED BEDS

258

▲ 2.6%

TOTAL STAFF ALLOCATION

47 Staff

HOSPITAL BED UTILIZATION

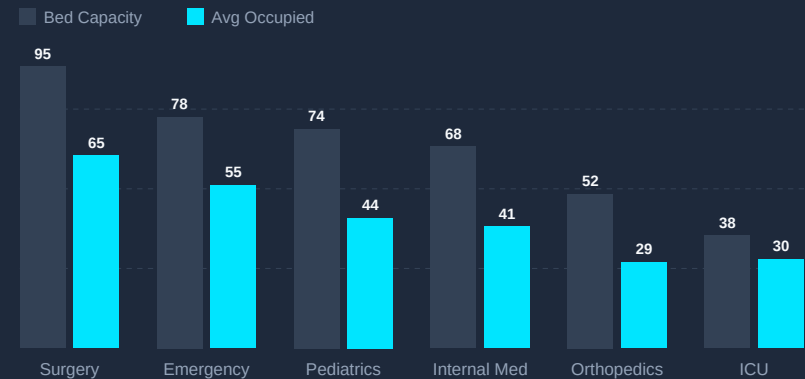
63.86%

▲ 1.1%

Total Bed Capacity vs. Avg Occupied Beds

Axis: Dept → Beds

Grouped Bar Chart



Staff Workload Ratio

Admissions per Staff Member

Horizontal Bar Chart



"WHAT-IF" CAPACITY SIMULATOR

Interactive Parameter Control — Volume Growth Projection

Simulate Volume Growth

Active: +10%



PROJECTED OCCUPANCY

70.19%

ICU STATUS

CRITICAL SURGE (88.0%)

Operational Alert & Action Matrix

Threshold Rules — Prescriptive Response Tiers

NORMAL

< 62%

Routine monitoring — no action required.

CAUTION

62% – 75%

Review staffing; monitor inflow trends closely.

SURGE PROTOCOL

≥ 75%

Activate surge staffing & overflow bed protocol.



Prescriptive Alert Engine: Automatically flags capacity bottlenecks and simulates surge scenarios in real time.